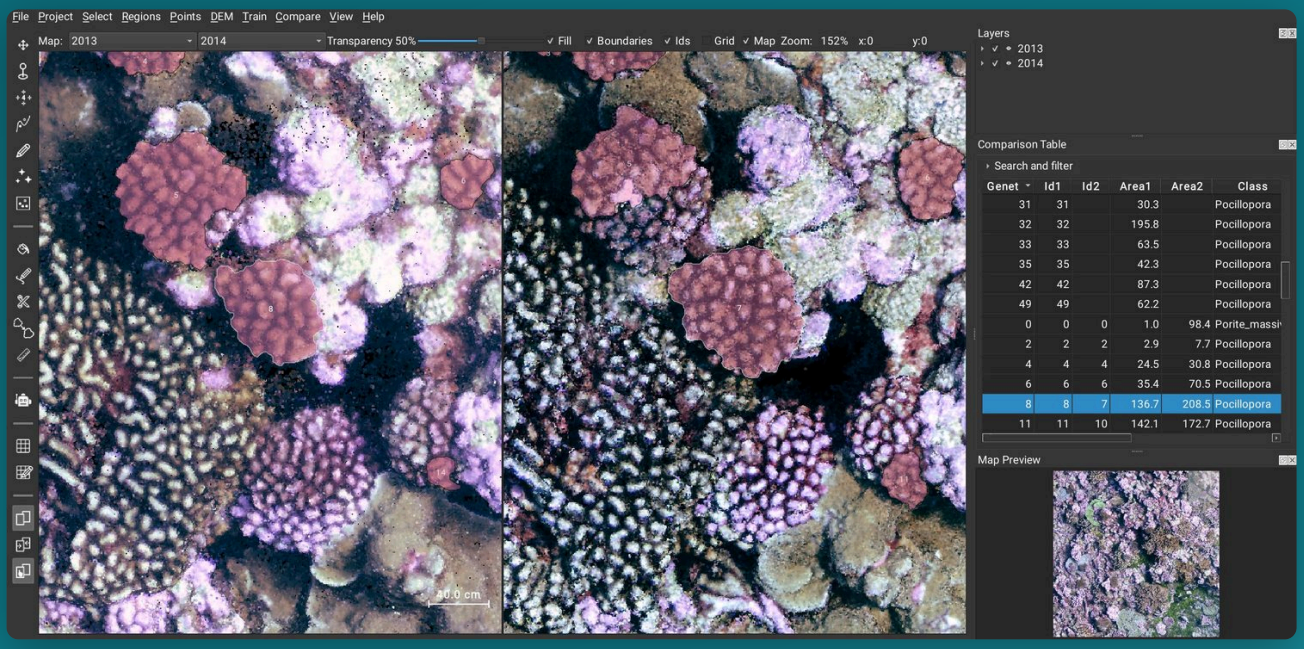
FIU CORAL REEF FISHERIES LAB · QUICK REFERENCE

TagLab Quick Reference

Tools, keys, menus and checklists for people who already annotate



Software: TagLab, FIU lab version (based on TagLab 2026.4.1, CNR-ISTI Visual Computing Lab)
Version 1.0 · November 2026 · New to TagLab? Read the *TagLab Guide for New Lab Members* first.
Screenshots: TagLab's official sample projects (CC BY-NC-SA 4.0; Stuart Sandin Lab, Scripps Institution of Oceanography; DEM sample courtesy of the Moorea Island Digital Ecosystem Avatar project).

Contents

1. The window and the basic rules 3
2. The toolbar 4
3. Tool cards 5
4. Selection, region operations and view keys 8
5. Checklists 9
6. Year-to-year at a glance 10
7. Exports and DEM metrics 11
8. Every menu command 12
9. Default keys 13
Glossary 14

1. The window and the basic rules

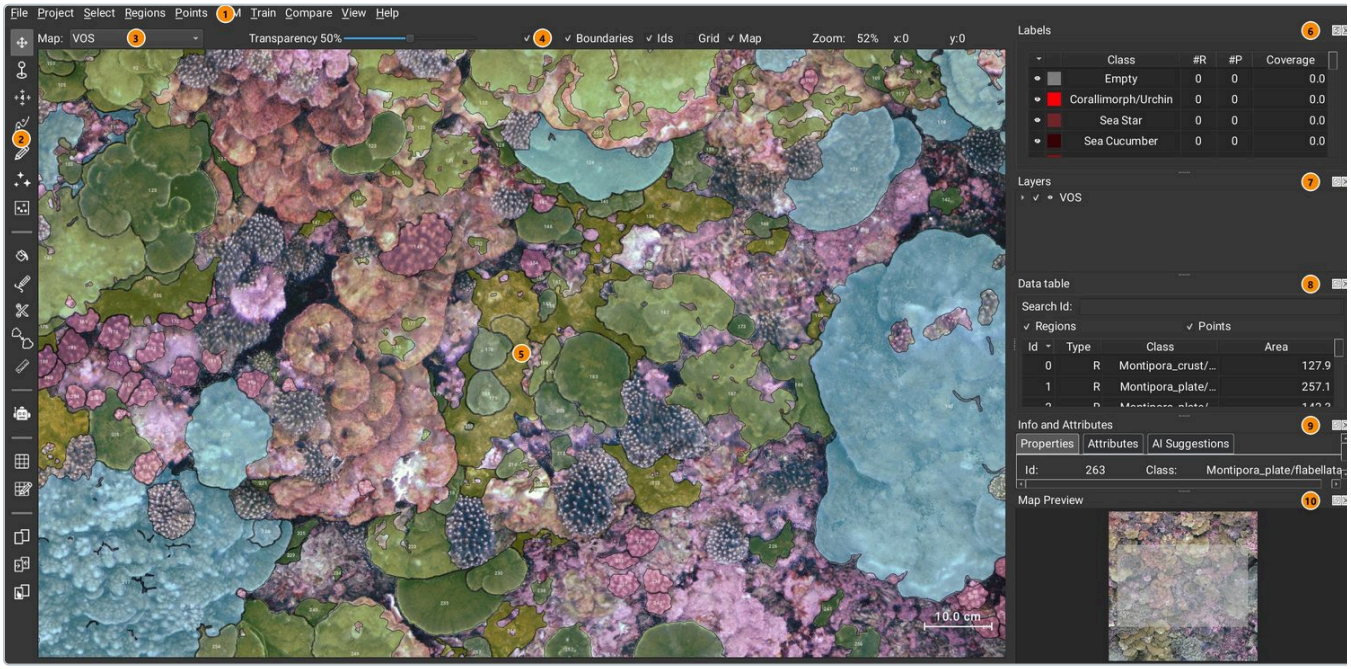


Figure 1. The TagLab window.

1. Menu bar
2. Toolbar (tools)
3. Map list (the survey shown)
4. Header bar: Fill, Boundaries, Ids, Grid, transparency, zoom, mouse position
5. Map viewer; the instructions of the current tool at its bottom left
6. Labels: classes, #regions, #points, area
7. Layers
8. Data table (comparison table in split screen)
9. Info and Attributes of the selected region
10. Map Preview: click or drag to move

The rules of every tool

- The mouse acts on the map **only with SHIFT held**: SHIFT + click (positive point, place, select), SHIFT + right click (negative point), SHIFT + drag (draw). A plain drag pans; Ctrl + drag pans in any tool.
- Space confirms, Esc cancels, Ctrl+Z removes the last point (or undoes the last operation), Ctrl+Shift+Z redoes.
- **Double-click** selects the region to work on (Edit Border, Cut).
- New regions are class **Empty**: give them a class.
- Points have **no undo**. Creating or deleting matches by hand has no undo.
- Save often (Ctrl+S) and turn on autosave (File ▸ Settings..).

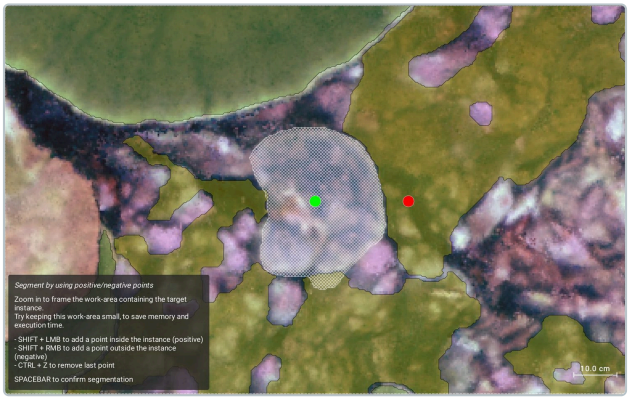
The view keys are in section 4.

2. The toolbar

	Button	Key	What it does
	Pan/Zoom	1	Pan and zoom the map; click a region to select it (SHIFT: add to the selection), SHIFT + drag: select with a rectangle.
	Place annotation point		Place an annotation point (SHIFT + click).
	4-clicks segmentation	2	Outline a colony from 4 clicks on its top, bottom, left and right edge (DEXTR).
	Positive/negative clicks segmentation	3	Outline a colony with positive (inside) and negative (outside) clicks (RITM). Also corrects a selected region.
	Freehand segmentation	4	Draw the outline by hand, then SPACE.
	Watershed segmentation (hidden here)		Scribbles inside and outside a colony (shown only in the Digital Heritage research field).
	SAM - all instances in an area		SAM: outline every object in a square work area (slow without an NVIDIA GPU).
	SAM - positive/negative clicks in an area		SAM with positive / negative clicks in a work area.
	Rows analysis (hidden here)		Rows analysis (Digital Heritage only).
	Assign class to region	5	Give the class chosen in the Labels panel to a region (SHIFT + click).
	Edit region border	6	Redraw part of the border of the selected region.
	Cut region	7	Cut the selected region in two along a line.
	Move region (SHIFT + drag or the arrow keys; SPACE re-fits the outline)		Move, turn and correct regions (drag, arrows, , and ., points).
	Measure tool	9	Measure a distance; set the pixel size from a known length.
	Fully auto semantic segmentation	0	Classify the map with a trained network (Select Classifier window).
	Create grid		Create a grid of cells on the map (to annotate cell by cell).
	Active/disactive grid operations		Turn grid operations on or off (cell states and notes).
	Split screen	Alt + S	Show two maps side by side (year-to-year).
	Compute automatic matches		Match the colonies of the two maps automatically.
	Edit matches		Edit the matches by hand.

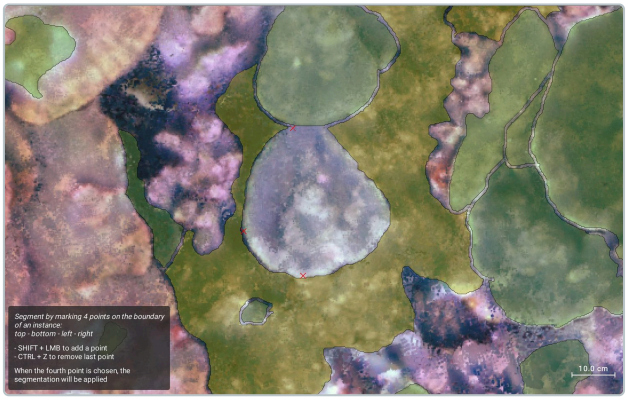
3. Tool cards

Positive/negative clicks (RITM) 3



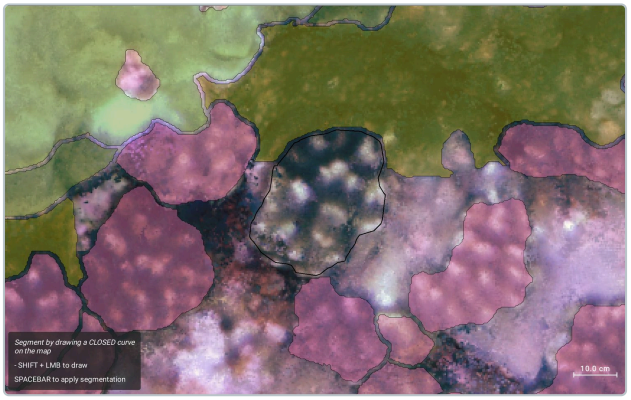
- Zoom on the colony first (the view is the work area).
- SHIFT + click inside, SHIFT + right click outside where it leaks.
- Space confirms. On a selected region: corrects it (keeps id, class, matches).

4 clicks 2



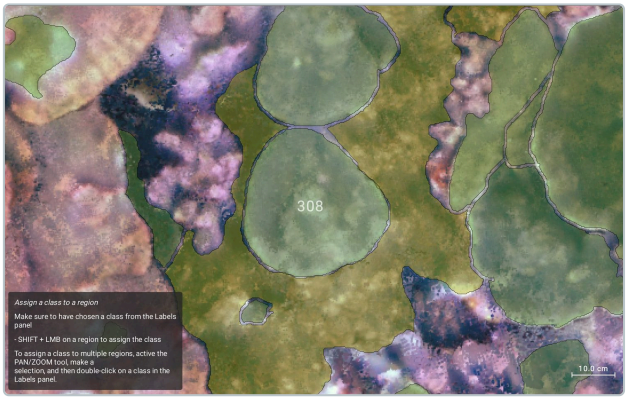
- SHIFT + click the top, bottom, left and right extreme points.
- The 4th click creates the region (no Space).

Freehand 4



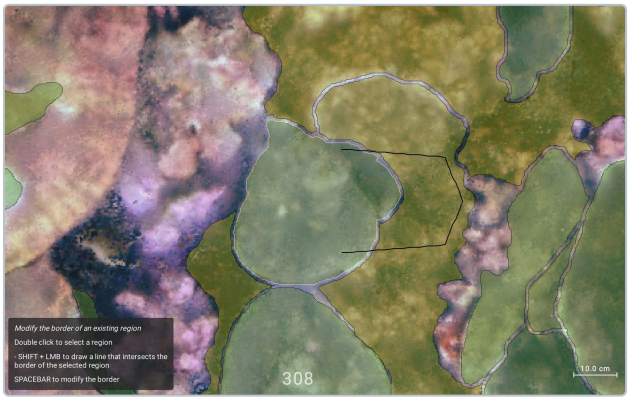
- SHIFT + drag a closed outline (several strokes allowed).
- Space creates the region.

Assign class 5



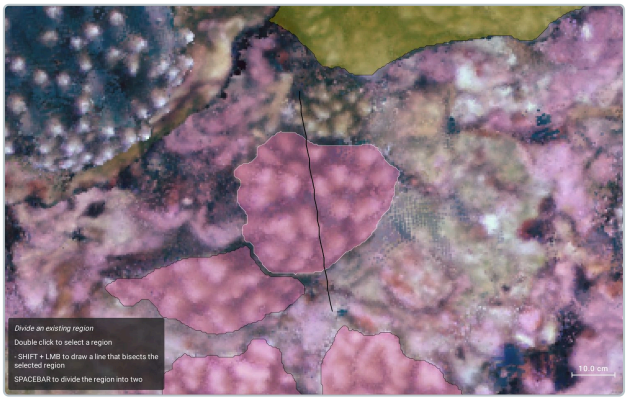
- Click the class in the Labels panel, SHIFT + click the regions.
- Many at once: select them, double-click the class (or A).

Edit Border 6



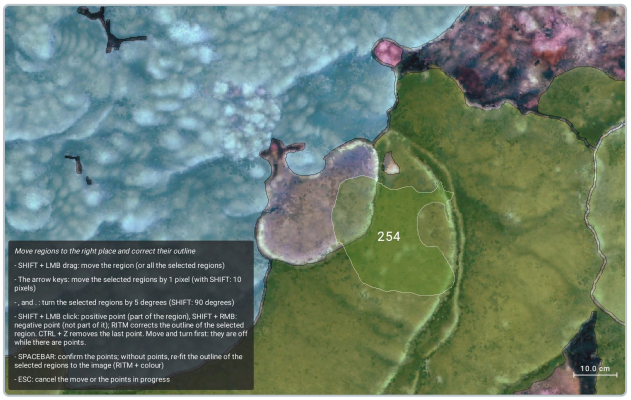
- Double-click the region.
- SHIFT + drag a line that crosses the border out and back in.
- Space applies.

Cut 7



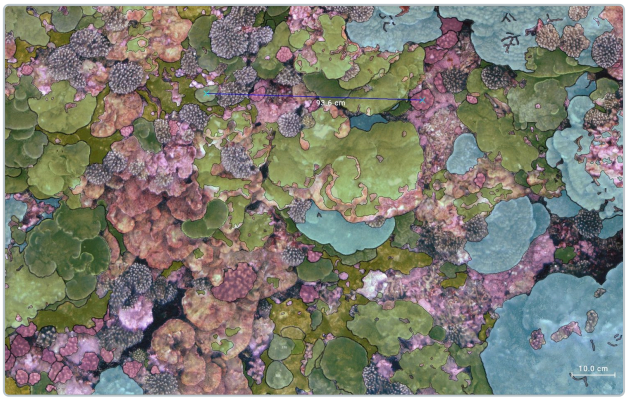
- Double-click the region.
- SHIFT + drag a line right across it, Space: two regions.

Move Region



- SHIFT + drag moves; arrows 1 px (SHIFT 10 px).
- , . turn 5°; SHIFT + , . turn 90°.
- Then SHIFT + click / SHIFT + right click points; Space confirms. Space without points re-fits the outline.

Ruler 9



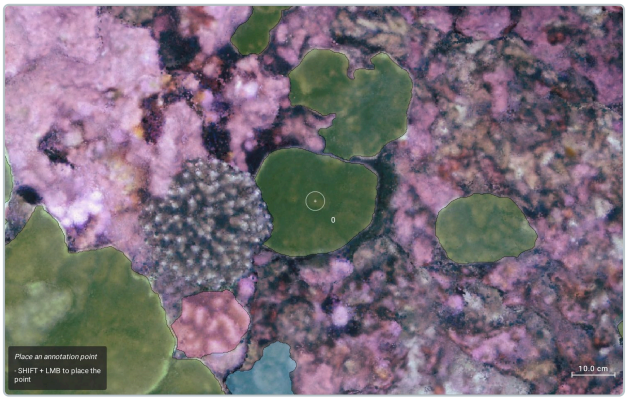
- SHIFT + click both ends: length in mm and px.
- Type the true length to set a new pixel size.

SAM



- SHIFT + wheel sizes the square, SHIFT + click places it; SHIFT + wheel sets the seed spacing; Space runs.
- GPU computer only. Check every outline.

Place point



- SHIFT + click places a point (Empty).
- Select points like regions; A gives the class; Del deletes.

4. Selection, region operations and view keys

Select	How
One / add or remove one	Click / SHIFT + click (Pan tool)
A rectangle	SHIFT + drag (Pan tool)
All / none / invert	Ctrl+A / Ctrl+D / Ctrl+I
By class, working area, properties	**Select** menu

Operation	Key	Needs	Result
Assign Class	A	any	Active class of the Labels panel
Delete	Del	any	Regions removed
Merge Overlapping Regions	M	2+	One region (union)
Divide Regions	D	2 overlapping	No more overlap
Subtract Regions	S	2	First minus second
Snap Borders	B	2 touching	Common border, no gap
Fill Region	F	any	Holes filled
Refine Border	R	1	Border onto the image edges
Dilate / Erode	+ / -	any	±1 pixel
Create a Background Region using the WA	N	working area	Empty regions where there is none

Regions ▸ **Refine All Borders** refines every region; **Regions** ▸ **Compute Geometric Info** measures the selected ones.

View	Key
Fill / boundaries / ids / grid on or off	Q / W / E / G
Image ↔ DEM	C
Grid cell state under the mouse	U
Split screen	Alt+S
Help (keys in use)	Ctrl+H
Zoom / pan	Mouse wheel / drag (Pan tool) or Ctrl + drag

5. Checklists

New plot (project lead)

☐ **File ▸ New Project**, then **Project ▸ Labels Dictionary Editor...** > Load the dictionary (FDEP / sponge nursery)

☐ **Project ▸ Add New Map...**: name `Plot_Year`, orthomosaic, DEM, date, pixel size

☐ Check the pixel size with the Ruler; check it in **Project ▸ Maps Editor...**

☐ **Project ▸ Region Attributes...** if the plot needs extra fields

☐ **Project ▸ Set Working Area**

☐ **File ▸ Save As** next to the images

Annotating a map

☐ Choose the map; check pixel size and working area

☐ Work in bands; grid on for large maps

☐ Outline (RITM first), give the class at once

☐ Ctrl + S every 10–15 min

☐ Fill off (Q); look for Empty regions and tiny regions (Data table sorted by Area)

☐ Save; note in the lab log what was done

DEM metrics

☐ Outlines finished (editing clears the values)

☐ Working area = the plot

☐ **Select ▸ Select None**, **DEM ▸ Calculate DEM Metrics**; read the message

☐ **DEM ▸ Calculate Surface Area** if needed

☐ Save; **File ▸ Export ▸ Export Annotations As Data Table**; **DEM ▸ Export Plot Metrics (CSV)**

☐ Compare years with `rugosity_1cm`, not `rugosity`

A new year

☐ **File ▸ Save As** a dated copy

☐ **Project ▸ Add New Map...**

☐ **Project ▸ Alignment Tool**: Auto Markers, untick bad markers, Preview, Check alignment, Confirm; save

☐ Split screen: old left, new right

☐ **Compare ▸ Carry Regions Over...**; review every check / possibly gone

☐ Check the ok ones; outline the recruits

☐ Comparison table: To check, Dead, Born

☐ **Compare ▸ Export Matches**; save

Training dataset → network → classification

☐ Annotated area complete; working area on it

☐ **Train ▸ Export New Training Dataset**: folder, split, target pixel size, *Merge classes*...

☐ Save the project (merges, new labels)

☐ On the GPU computer: **Train ▸ Train Your Network**; read the per-class IoU; Confirm or Cancel

☐ Classifier window (0): Preview a small area, threshold, Apply

☐ Check and correct every region

CoralNet training points

☐ Save the project (points have no undo)

☐ **Points ▸ Sample Points On This Map** (optional)

☐ **Points ▸ Label Points from Regions...**: min points per class, edge distance; Yes

☐ **Points ▸ Export Tiled Data for CoralNet**: tile size, codes

☐ Upload to a **test** source first (website or Toolbox)

6. Year-to-year at a glance

Fate	Meaning
grow	area2 > 1.05 × area1
shrink	area2 < area1 / 1.05
same	in between
dead	no match in the newer map, or set by hand (double-click Action)
born	no match in the older map
split / fuse	1 → several / several → 1 (the group's fate)
dark-red row	to check

Carry-over	Meaning
`ok`	Found in place, outline fitted
`check`	Doubtful (the *Why* column says why)
`possibly gone`	Probably dead or missing
`kept`	Checked and kept in the review

Review	Key
Next / previous	N / P
Keep	K
Remove (→ dead)	Del
Move / correct, Re-fit outline	buttons

Auto-match: same class, not Empty, overlap ≥ 60% of the smaller colony.

Alignment Tool: SHIFT + click the same spot on both maps adds a marker (SHIFT + click on a marker: soft ↔ hard); right drag moves it; Del deletes it. Untick markers with large errors. Check alignment: green ≤ 1 mm, yellow ≤ 3 mm, red > 3 mm, grey not measurable. Confirm replaces the map (files `_coreg`), keeps its name and id.

Edit matches (): SHIFT + click the colony on each map, Space creates the match; select a row of the comparison table and Del deletes it.

7. Exports and DEM metrics

Export	Menu	Needs
Data table (CSV)	**File ▸ Export ▸ Export Annotations As Data Table**	Visible classes only; working area if set
Shapefile	**File ▸ Export ▸ Export Regions As Shapefile**	Georeferenced map
Label image	**File ▸ Export ▸ Export Regions As Labeled Image (PNG)**	—
Label / ortho GeoTIFF	**File ▸ Export ▸ Export Regions As A GeoTiff**, **File ▸ Export ▸ Export Orthoimage As A GeoTiff**	Georeferenced map
Size histogram	**File ▸ Export ▸ Export Histogram From Regions**	—
Plot DEM metrics	**DEM ▸ Export Plot Metrics (CSV)**	DEM metrics computed
Clipped DEM	**DEM ▸ Export Clipped Raster**	DEM
Matches (tidy CSV)	**Compare ▸ Export Matches**	Comparison table
Genets (long CSV)	**Compare ▸ Export Genet Data as CSV**	Comparison table
Training dataset	**Train ▸ Export New Training Dataset**	—
CoralNet training data	**Points ▸ Export Tiled Data for CoralNet**	Labelled points

Colony DEM metric	Unit	Meaning
`slope_deg`	°	Mean slope (walls > 87° left out)
`elev_mean/min/max`	m	Elevation
`relief_cm`	cm	Highest − lowest point
`height_cm`	cm	Top above the plane of the substrate around it
`volume_cm3`	cm^3	Volume above that plane
`rugosity`	—	Surface / planar area

Units: area cm^2 = pixels × s^2 / 100, perimeter cm = pixels × s / 10, with s the pixel size in mm.

8. Every menu command

File

- New Project Ctrl + N
- Open Project Ctrl + O
- Save Project Ctrl + S
- Save As Ctrl + Alt + S
- Import
  - Import Label Image
  - Import Shapefile
  - Add Another Project
- Export
  - Export Annotations As Data Table
  - Export Regions As Labeled Image (PNG)
  - Export Regions As Shapefile
  - Export Regions As DXF
  - Export Regions As SVG
  - Export Regions As A GeoTiff
  - Export Orthoimage As A GeoTiff
  - Export data for Reef Archive Lab platform
  - Export Histogram From Regions
  - Export New Training Dataset
- Settings..
- Keyboard and Mouse Shortcuts...

Project

- Add New Map... Ctrl + L
- Maps Editor... Ctrl + M
- Georeferencing Tool
- Set Working Area
- Alignment Tool
- Labels Dictionary Editor...
- Region Attributes...

Select

- Select All Ctrl + A
- Select None Ctrl + D
- Invert Selection Ctrl + I
- Select by current Class
- Select by Working Area
- Select by Properties

Regions

- Undo Ctrl + Z
- Redo Ctrl + Shift + Z
- Assign Class to Region A
- Delete Region Del
- Merge Overlapping Regions M
- Divide Regions D
- Subtract Regions S
- Snap Borders B
- Fill Region F
- Refine Border R
- Refine All Borders
- Dilate Border +
- Erode Border -
- Create a Background Region using the WA N
- Compute Geometric Info
- Course Analysis

Points

- Sample Points On This Map
- Label Points from Regions...
- Import Viscore Point Annotations
- Import CoralNet Point Annotations
- Export CoralNet Point Annotations
- Export Tiled Data for CoralNet
- Open CoralNet-Toolbox...

DEM

- Switch RGB/DEM
- Calculate Surface Area
- Calculate DEM Metrics
- Export Plot Metrics (CSV)
- Export Clipped Raster

Train

- Export New Training Dataset
- Dataset Manager
- Train Your Network

Compare

- Enable Split Screen Alt + S
- Compute Automatic Matches
- Edit Manual Matches
- Export Matches
- Export Genet Data As Shapes
- Export Genet Data as CSV
- Carry Regions Over...
- Review Carried-over Colonies
- Clear Comparison Table

View

- Labels
- Layers
- Info and Attributes
- Map Preview
- Data table

Help

- Help Ctrl + H
- Learn TagLab
- Report Issues
- About TagLab

9. Default keys

Keys can be changed, and mouse buttons added, with **File ▸ Keyboard and Mouse Shortcuts...**.

File and edit

New project	Ctrl + N
Open project	Ctrl + O
Save project	Ctrl + S
Save project as	Ctrl + Alt + S
Add new map	Ctrl + L
Maps editor	Ctrl + M
Help (list of the main keys)	Ctrl + H
Undo	Ctrl + Z
Redo	Ctrl + Shift + Z
Select all	Ctrl + A
Select none	Ctrl + D
Invert selection	Ctrl + I

Regions

Assign class to region	A
Delete region	Del
Merge overlapping regions	M
Divide regions	D
Subtract regions	S
Snap borders	B
Fill region	F
Refine border	R
Dilate border	+
Erode border	-
Create a background region using the working area	N

View

Split screen on / off	Alt + S
Switch RGB / DEM	C
Show / hide the fill of the regions	Q
Show / hide the borders	W
Show / hide the region ids	E
Show / hide the grid	G
Grid: change the state of the cell under the mouse	U

Tools

Pan / zoom	1
4-clicks segmentation	2
Positive / negative clicks segmentation (RITM)	3
Freehand segmentation	4
Assign class (tool)	5
Edit region border	6
Cut region	7
Create crack	8
Measure tool (ruler)	9
Fully automatic segmentation (classifier)	0
Move region	none
Place annotation point	none
Watershed segmentation	none
SAM: positive / negative clicks in an area	none
SAM: all instances in an area	none
Edit matches	none

Confirm and cancel

Confirm the operation of the tool	Space
Cancel the operation of the tool	Esc

Carry-over review

Next colony	N
Previous colony	P
Keep the colony	K
Remove the colony	Del

Move Region tool

Move left (SHIFT: 10 pixels)	Left
Move right (SHIFT: 10 pixels)	Right
Move up (SHIFT: 10 pixels)	Up
Move down (SHIFT: 10 pixels)	Down
Turn 5 degrees counter-clockwise (SHIFT: 90 degrees)	,
Turn 5 degrees clockwise (SHIFT: 90 degrees)	.
Turn 90 degrees counter-clockwise	<
Turn 90 degrees clockwise	>

Glossary

Maps and projects

Project
A TagLab `.json` file. It holds the list of maps (with the paths of their image files), the labels, every region and point, the comparison tables and the settings of the project. The images are not inside it: keep the images where the project expects them, or TagLab asks for them when it opens the project.

Map
One survey of one plot in a project: an orthomosaic, optionally with its DEM, a name, a pixel size and an acquisition date. Called *image* in some TagLab messages.

Plot
The area of reef surveyed and compared over time (for example a Culebra year-to-year plot or a PEV plot).

Orthomosaic (ortho)
A single top-down image of the plot, stitched from many photos by photogrammetry (Metashape, Viscore…), with the same scale everywhere.

DEM (digital elevation model)
A raster of heights (one value per pixel, in metres) made with the orthomosaic. TagLab uses it for surface area, DEM metrics, plot rugosity and the height checks of the Alignment Tool and Carry Regions Over.

Channel
One image of a map: **RGB** (the orthomosaic) or **DEM**. C switches the view between them.

Pixel size
The size of one map pixel on the reef, in millimetres (mm/px). Every area, perimeter and length in cm depends on it. TagLab never reads it from the GeoTIFF: type it in Map Settings or measure it with the Ruler.

Acquisition date
The survey date of a map (YYYY-MM-DD). The year-to-year exports use it to compute the time between two maps.

Georeference
The position of the map in real-world coordinates, stored in a GeoTIFF. Needed by the shapefile and GeoTIFF exports; when both maps of a pair have one, it is also used to line them up.

GeoTIFF
A TIFF image with a georeference. Orthomosaics and DEMs exported by photogrammetry software are usually GeoTIFFs.

Working area
A rectangle of the map set with Project ▸ Set Working Area. Exports, DEM plot metrics, Label Points from Regions and the training dataset can be limited to it. Not the same as the *work area* of RITM or SAM.

Work area (RITM, SAM)
The part of the map an AI tool looks at: for RITM, the view at the first click; for SAM, the square chosen in step 1. Keep it small: larger areas are slower and less precise.

Grid
A grid of cells drawn on the map (toolbar button *Create grid*), to annotate a large map cell by cell. Each cell can be marked empty, incomplete or complete, with a note.

Sampling area
An area where Points ▸ Sample Points On This Map places points (a single rectangle, the working area, or transects).

Layer
An extra drawing over a map (for example an imported shapefile, the grid or sampling areas), listed in the Layers panel.

Label, class
The category a region or point belongs to (a species, Sand, Brick…), with a fill colour. The project's list of labels is shown in the Labels panel.

Dictionary
A `.json` file with a list of labels and their colours (for example `FDEP.json`, `spongenursery.json`). It is copied into the project when you load it; later changes to the file are not synced.

FDEP list
The Florida Department of Environmental Protection species list used by the lab (`FDEP.json`).

Region attributes
Extra fields that every region can have (numbers, text or keyword lists), defined in **Project ▸ Region Attributes...**. DEM metrics and the carry-over status are stored as region attributes.

Autosave file
`<project>_autosave.json`, written next to the project every few minutes when autosave is on. It never replaces the project file.

`TagLab.log`
The log file in the TagLab folder. Errors, timings and the results of long operations are written there.

`config.json`
TagLab's settings file in the TagLab folder. It lists the available classifiers (networks).

Regions and annotation

Region
One outlined object on a map (a colony, a sponge, a patch of sand...): an outline, optional holes, a class, an id and measurements. Called *blob* in the code.

Outline (contour)
The border of a region. Holes inside a region are *inner contours*.

Id
The number of a region in its map. Ids are never reused, so a deleted id stays unused.

Empty
The class of a new region before a class is assigned. Empty regions are skipped by auto-match and by most exports.

Point (annotation point)
A single annotated position with a class, as in point-count surveys and CoralNet.

Selection
The regions (and points) currently highlighted. Operations such as Merge, Delete or Calculate DEM Metrics work on the selection.

Active map
In split screen, the map you clicked last. Menu commands work on it.

Split screen
Two maps side by side (usually two years of the same plot), each with its own view.

Tool instructions
The panel at the bottom left of the map that lists the mouse and keys of the current tool.

Positive / negative point
A click that tells an AI tool "this is part of the object" (positive, SHIFT + left click) or "this is not" (negative, SHIFT + right click).

RITM
The positive/negative clicks tool: an AI model that outlines an object from a few clicks. It also corrects an existing region. Works well on a CPU.

DEXTR (4 clicks)
An AI model that outlines an object from its four extreme points (top, bottom, left, right).

SAM (Segment Anything)
A large AI model that outlines every object in an area, or one object from clicks. Very slow without an NVIDIA GPU, and optional at install time.

Refine Border
Moves the border of the selected region onto the nearest strong edges of the image.

Snap Borders
Makes the borders of touching regions follow each other exactly (no gaps or overlaps).

Dilate / Erode
Grow or shrink the selected regions by one pixel all around.

Background region
New regions (class Empty) that cover every part of the working area without a region
(**Regions ▸ Create a Background Region using the W**
Give them a class such as Sand.

Background (training)
In a training dataset, the pixels that belong to no exported class (black in the label images).

Measurements

Area
The planar (top-down) area of a region in cm^2: pixels × (pixel size)^2 / 100.

Perimeter
The length of the outline in cm: pixels × pixel size / 10.

Surface area
The 3D area of the region measured on the DEM (always larger than the planar area). Computed with DEM ▸ Calculate Surface Area.

Slope
The steepness of the reef surface, in degrees from horizontal, computed from the DEM with Horn's method (as `gdaldem slope`).

Rugosity
Surface area divided by planar area: 1 for a flat surface, larger for a rougher one. The value depends on the DEM resolution, so compare plots with `rugosity_1cm`.

`rugosity_1cm`, `slope_1cm`
Plot rugosity and slope computed on the DEM averaged to 1 cm cells. Use them to compare surveys whose DEMs have different resolutions.

Relief
For a colony: the height range inside its outline (cm). For the plot: the height range after removing the plot's overall tilt (1st to 99th percentile).

Height above plane, volume
The height (cm) and volume (cm^3) of a colony above a plane fitted to the substrate just outside its outline.

Elevation
The DEM value (metres) inside a region: `elev_mean`, `elev_min`, `elev_max`.

DEM metrics
The values stored per region by DEM ▸ Calculate DEM Metrics: `slope_deg`, `elev_mean`, `elev_min`, `elev_max`, `relief_cm`, `height_cm`, `volume_cm3`, `rugosity`. Regions with less than half of their area covered by the DEM get no values.

Plot metrics
The rugosity, slope and relief of the whole plot (the working area or the whole map), stored with the map and exported by DEM ▸ Export Plot Metrics (CSV).

Centroid, bounding box
The centre of a region, and the smallest rectangle that contains it (in map pixels).

Year-to-year

Timepoint
One survey of a plot (one map in the project).

Alignment (co-registration)
Transforming the new map so that it lies exactly on the reference map, pixel for pixel. Done with the Alignment Tool before comparing years.

Reference map, aligned map
In the Alignment Tool, the map that stays as it is (reference, left) and the map that is moved onto it (right).

Marker
A pair of points, one on each map, on the same feature. The alignment is fitted to the markers. A *hard* marker (red; SHIFT + click on a marker toggles it) counts three times as much as a *soft* one in the fit.

Auto Markers
Finds markers automatically by matching image features (and, with DEMs, the relief), whatever the rotation.

RMSE
Root mean square error: the typical distance (mm) between where the markers are and where the fitted alignment puts them. Small is good; a marker with a large error is shown in red.

Apply scale
Lets the alignment also change the size of the moved map (when the two surveys have slightly different scales). Off by default (rotation and shift only).

Check alignment, quality map
A measurement, in the Alignment Tool preview, of the misalignment left after the fit, on a grid of cells: green ≤ 1 mm, yellow ≤ 3 mm, red > 3 mm, grey not measurable.

Comparison table
The table of matches between two maps (one table per pair), shown in split screen in place of the data table.

Match
A link between a region of the first map and a region of the second map: the same colony in the two surveys.

Fate
What happened to a colony between the two maps, in the Action column: **grow** / **shrink** (area changed by more than 5%), **same**, **dead** (no match in the second map, or set by hand), **born** (no match in the first map). **n/s** (not set) appears only for a moment, before the areas are computed.

Split, fuse
One colony matched to several (split) or several matched to one (fuse). The whole group gets one fate.

Genet
TagLab's lineage number: all the regions linked by matches across the maps form one genet. It is recomputed when matches change.

Colony name
A stable name for a colony in the exports, `map:id` of its first region (for example `2013:45`).

To check
A match TagLab could not keep with certainty (after an edit, a repair or a doubtful carry-over), shown in dark red.

Placement
How Carry Regions Over and auto-match line up two maps: *same grid*, *georeference*, or an *offset* found from the images.

Carry-over
Copying the regions of an annotated map onto the next survey (Compare ▸ Carry Regions Over...), refitting their outlines and checking whether each colony is still there.

Carry-over status
The `carryover` attribute of a carried region: **ok**, **check** (look at it), **possibly gone**, or **kept** (confirmed in the review). `carry_conf` is the confidence (0–1) and `carry_from` the id on the old map.

AI

Training dataset
Image tiles and their label images exported from annotated maps (Train ▸ Export New Training Dataset), split into training, validation and test sets.

Tile
A square piece of the map (513 × 513 pixels for TagLab's networks) used for training and classification.

Target pixel size
The pixel size the tiles are resampled to. A classifier works best on maps with the pixel size it was trained on.

Dataset split
How the map is divided into training, validation and test areas: *uniform* (vertical or horizontal strips) or *biologically-inspired* (areas chosen so that each set has a similar mix of classes).

Training / validation / test set
Training tiles teach the network; validation tiles choose the best epoch; test tiles give the final, independent score.

`classes.json`
The file in an exported dataset that lists each class, its colour and the classes it was merged from. Train Your Network reads the classes from it.

Class merge
Exporting several classes under one name (for example "Vase (Boat)" + "Vase (Scuba)" → "Vase") without changing the regions.

Network, classifier (`.net`)
A trained model saved in `models/`. Classifiers are listed in `config.json` and chosen in the Select Classifier window.

DeepLab V3+
The type of semantic segmentation network TagLab trains (with a ResNet-101 backbone).

Semantic segmentation
Giving a class to every pixel of the map. TagLab then turns the classified pixels into regions.

Epoch
One pass of the training over all the training tiles.

Batch size
How many tiles the network sees at once. Larger batches need more GPU memory.

Learning rate
How large the steps are when the network's weights are adjusted.

L2 regularization
A penalty on large weights that reduces overfitting.

Preset
Training schedule: *Preset 1* adjusts all the weights of the network; *Preset 2* trains the last layer first, then the decoder, then the whole network.

Loss
The training error the network minimises. It should go down over the epochs for both training and validation.

Accuracy
The fraction of pixels classified correctly. It is dominated by the common classes (often the background).

IoU, mIoU
Intersection over union: the overlap between predicted and true pixels of a class, divided by their union (1 = perfect). mIoU is the mean over the classes (Background excluded); TagLab keeps the epoch with the best mIoU.

Confusion matrix
A table of how many pixels of each true class were predicted as each class.

Overfitting
When the network learns the training tiles by heart and does worse on new ones: training loss keeps going down while validation loss goes up.

Augmentation
Random changes to the training tiles (flips, rotations, colour changes, small holes) so that the network generalises better.

Ignore unannotated pixels
A training option: pixels with no label are left out instead of being taught as background. Use it only for areas annotated completely.

Preview
Classifying a small area in the Select Classifier window, to check a network before classifying the whole map.

Fast / Maximum quality
The two classification modes: overlapping tiles with smooth blending (default), or the old 9-pass method (about 8 times slower, nearly the same result).

CPU, GPU, CUDA
The processor and the graphics card. Training and SAM need an NVIDIA GPU (CUDA) to be practical; annotation tools such as RITM run fine on the CPU.

`batch_processing.py`
A script that classifies all the projects of a folder with a classifier, without the window (for the GPU PC).

CoralNet, source, label set
CoralNet is the online point-annotation and machine-learning platform. A *source* is a CoralNet project; its *label set* lists the labels it uses, each with a short *label code*.

CoralNet Toolbox
The window (Points ▸ Open CoralNet-Toolbox...) that signs in to CoralNet and uploads tiles and annotations or gets predictions.

Software

`venv`
TagLab's own Python environment, in the `venv` folder of the TagLab folder.

`opentaglab.bat`
Starts TagLab (the desktop shortcut points to it).

`install_windows.bat`
Installs the Python packages, PyTorch (GPU version when there is an NVIDIA card) and the AI models.

`update.bat`
Updates TagLab from the lab's GitHub repository. Replaced files are saved first in `update_backups\`.

`download_models.bat`
Downloads missing or incomplete AI models; `--check` only lists them.

Shortcut editor
File ▸ Keyboard and Mouse Shortcuts...: changes any key, or puts an action on a mouse button.